

HW

Find the frequency counts for the following

```
for (i=1; i<n; i++);  
printf("Hello");
```

```
1) i=1;  
   i=i+20;  
   printf("%d", i);
```

```
3) i=1;  
   while (i<=n)  
   {  
       printf("%d", i); i++;  
   }
```

4) For all the shell programs done in OS lab slides find out the time complexity in terms of frequency count.

1. `for(i=1; i<n; i++)
printf("Hello");`

Statement	Frequency count
<code>i=1</code>	1
<code>i<n</code>	n
<code>i++</code>	n-1
<code>printf("Hello")</code>	1

Time complexity: $O(n)$

2. `i=1;
i=i+20;
printf("%d", i);`

Statement	Frequency count
<code>i=1</code>	1
<code>i=i+20</code>	1
<code>printf("%d", i)</code>	1

Time Complexity: $O(1)$

3. `i=1;
while (i<=n)
{
printf("%d", i);
i++;
}`

Statement

Frequency Count

$i = 1$

1

$i \leq n$

$n+1$

printf ("%d", i)

n

$i++$

n

Time complexity : $O(n)$

4) Addition of two numbers

Statement

Frequency Count

echo - n "Enter
the first number"

1

read num 1

echo - n "Enter
the second number"

read num 2

sum = \$((num 1 + num 2))

echo "The sum of \$num 1 &
\$num 2 is: \$sum"

Total frequency count

$T(n) = 1 + 1 + 1 + 1 + 1 + 1 = 6$

$T(n) = 6$

$O(1) = \text{Time complexity}$

Menu driven Script

Statement	Frequency
Variable Initialization	4
Display menu	6
Read user's choice	1
case \$choice in	1
Selected case execution	2
exit(done)	1

Total frequency count

$$T(n) = 4 + 6 + 1 + 1 + 2 + 1 = 15$$

$T(n) = 15$ = Time complexity.

Shell script program

Statement	Frequency
echo(prompt)	1
$a = \$(wc -l < count)$	1
$b = \$(wc -m < count)$	1
$c = \$(tr -cd 'a-zA-Z0-9' < count) wc -m$	1
$d = \$(tr -cd " " < count) wc -m$	1

Statement	Frequency
echo "Total no. of lines: "\$a	1
echo "Total no. of character: "\$b	1
echo "Total no. of vowels: "\$c	1
echo "Total no. of blank spaces: "\$d	1

Total frequency count

$$T(n) = 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 = 9$$

$$T(n) = 9$$

$$\text{Time complexity} = O(1)$$